# AMIRREZA ASADI

**PhD Candidate in Mechanical Engineering | Computational Biomechanics**

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## RESEARCH PROFILE

PhD candidate developing computational and physics-informed methods to identify nonlinear soft-tissue properties from deformation and medical-imaging data. Experience spans hyperelastic constitutive modeling, inverse problems, optimal experimental design, finite-element simulation, full-field displacement analysis, and deep learning. Research emphasizes identifiable, repeatable, and spatially resolved mechanical characterization.

## EDUCATION

**PhD in Mechanical Engineering** | University of California, Riverside — **2023-Present**

*Advisor: Kaveh Laksari, PhD | Laksari Lab* | Riverside, California

Research focus: computational biomechanics, nonlinear tissue mechanics, inverse characterization, medical imaging, and physics-informed machine learning.

**MS in Mechanical Engineering** | University of California, Riverside — **2024**

*Graduate training in computational and solid mechanics* | Riverside, California

**BS in Mechanical Engineering** | Sharif University of Technology

*GPA: 17.62/20 | Graduated among the top students in the cohort* | Tehran, Iran

## RESEARCH EXPERIENCE

**Graduate Student Researcher** | Laksari Lab, University of California, Riverside — **2023-Present**

*Computational Biomechanics and Inverse Problems* | Riverside, California

- Develop inverse computational frameworks for estimating spatially varying hyperelastic material parameters from three-dimensional deformation fields and controlled loading data.
- Designed a stress-constrained physics-informed U-Net that predicts voxel-wise Mooney-Rivlin parameters while incorporating equilibrium residuals and boundary-traction consistency.
- Built finite-element and synthetic-data pipelines for heterogeneous volumetric samples across multiple loading, displacement-noise, and spatial-smoothing conditions.
- Developed an optimal experimental design and identifiability framework to select informative mechanical tests and improve repeatability of hyperelastic parameter estimation.
- Investigate noise-aware constitutive-model compatibility, atlas-guided model discrimination, medical-image registration, and ultrasound-based deformation reconstruction for experimental biomechanics.

## PEER-REVIEWED PUBLICATIONS

1. **Amirreza Asadi** and Kaveh Laksari. "Stress-Constrained Physics-Informed UNet for Voxel-Wise Multiparameter Hyperelastic Inversion in Volumetric Medical Imaging." *Annals of Biomedical Engineering (2026).* doi:10.1007/s10439-026-04321-4
2. **Amirreza Asadi** and Kaveh Laksari. "Optimal Experimental Design for Repeatable Hyperelastic Material Characterization." *Journal of the Mechanical Behavior of Biomedical Materials, 170, 107104 (2025).* doi:10.1016/j.jmbbm.2025.107104

## SELECTED RESEARCH PROJECTS

**Noise-Aware Atlas for Hyperelastic Model Identification** **2025-Present**

Developing a high-dimensional response atlas to distinguish, compare, and quantify compatibility among nonlinear constitutive models under realistic experimental noise.

**Methods:** Hyperelastic modeling, direct parameter fitting, probabilistic compatibility, manifold/atlas construction, topology validation, GPU-accelerated Python.

**Physics-Informed Volumetric Hyperelastic Inversion** **2024-2026**

Created a voxel-wise inverse framework that recovers multiple constitutive parameters from volumetric displacement data while enforcing mechanical consistency.

**Methods:** PyTorch, physics-informed loss design, equilibrium and traction constraints, finite-element simulation, synthetic 3D medical-imaging data.

**Optimal Experimental Design for Constitutive Characterization** **2023-2025**

Formulated a systematic framework for identifying informative loading protocols and reducing parameter uncertainty in nonlinear soft-tissue characterization.

**Methods:** Finite-element analysis, identifiability, sensitivity analysis, optimization, full-field deformation data, hyperelastic parameter estimation.

**Ultrasound-Based Deformation and 3D Reconstruction** **2026-Present**

Exploring reproducible ultrasound sweep acquisition, image registration, and displacement-field reconstruction for mechanically loaded soft-tissue phantoms.

**Methods:** DICOM processing, image registration, 2D displacement estimation, robotic sweep concepts, volumetric reconstruction.

## HONORS AND AWARDS

**Outstanding Poster Presentation Award** | Annual Research Symposium, University of California, Riverside **2026**

*Recognized for excellence in research communication and poster presentation*

## TECHNICAL EXPERTISE

**Computational biomechanics:** Nonlinear continuum mechanics, hyperelastic constitutive models, soft-tissue mechanics, finite-element methods, boundary-value problems.

**Inverse methods and statistics:** Parameter identification, optimization, optimal experimental design, sensitivity and identifiability analysis, uncertainty/noise evaluation, model compatibility.

**Programming and machine learning:** Python, PyTorch, NumPy, SciPy, pandas, GPU/CUDA-enabled scientific computing, physics-informed neural networks, convolutional neural networks.

**Simulation and imaging:** COMSOL Multiphysics, Blender-based geometry/data generation, DICOM processing, medical-image registration, full-field displacement analysis, digital image correlation concepts, 3D reconstruction.

**Research computing:** Google Colab, Jupyter, Git/GitHub, reproducible scientific pipelines, numerical validation, scientific visualization.

## RESEARCH SOFTWARE AND OPEN SCIENCE

**Optimal Hyperelastic Parameter Identification -** github.com/klaksari/optimal_hyperelastic_parameter_ID. Code associated with the 2025 optimal experimental design publication.

## RESEARCH INTERESTS

Computational biomechanics; nonlinear and heterogeneous tissue mechanics; hyperelasticity; inverse problems; elastography; medical imaging; physics-informed machine learning; optimal experiment design; model selection and identifiability; full-field deformation measurement.

*Last updated: August 2026*